

Activity Decision Making

ITA0506 - Computer vision

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Course: Computer vision

Code: ITA0506

1. Medical Image Enhancement:

understanding of problem content:

An x-ray image clearly shows the bones, but the darker soft tissues have poor visibility because their pixel intensities are concentrated in the enhance. the visibility of these darker regions without losing important details.

application of concepts:

1. which gray level transformation would you recommend?
log transformation

2. why is it suitable?

It enhances dark tissues by increasing the brightness of low intensity pixels while keeping bones visible.

3. what effect will it have on the image:

soft tissues become clear. contrast improves, and the x-ray is easier to analyse.

2. Night - Time surveillance.

A CCTV camera captures a parking lot at night. most pixel values are concentrated between 0 and 100, making the image very dark.

1. which transformation would improve the visibility?

log transformation.

2. why is it more suitable than negative transformation?

log transformation brightens dark regions and reveals hidden details, whereas negative transformation only inverts the image.

3. predict the appearance of the enhanced image.

The image appears brighter, making vehicles, people, and other objects easier to see.

3. Fingerprint Recognition

A fingerprint image contains many fine ridge details, but the contrast between ridges and valleys is very low.

1. Which gray level transformation would you apply?
Contrast stretching

2. Explain why.

It increases the contrast between fingerprint ridges and valleys, making fine details clear.

3. What would happen if a log transformation were applied instead?

The image would become brighter but ridge details may become less distinct.
summary.